

Rapid #: -25643080

CROSS REF ID: **6571592**

LENDER: **GAT (Georgia Institute of Technology) :: Main Library**

BORROWER: **HUL (Harvard University) :: Widener Library**

TYPE: Article CC:CCL

JOURNAL TITLE: Annales geophysicae. Series B, Terrestrial and planetary physics

USER JOURNAL TITLE: Annales geophysicae. Series B, Terrestrial and planetary physics

ARTICLE TITLE: Inversion of profiles of uplift rates for the geometry of dipslip faults at depth, with examples from the Alps and Himalaya

ARTICLE AUTHOR: MOLNAR P.

VOLUME: 5

ISSUE:

MONTH:

YEAR: 1987

PAGES:

ISSN: 0980-8760

OCLC #:

PATRON NOTES: 5

Processed by RapidX: 11/14/2025 1:55:58 PM

This article has been provided to you by the Information Delivery Department at the Georgia Tech Library. If you require any accessibility remediation, or have any other questions or concerns, please contact us at delivery@library.gatech.edu.

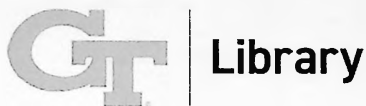
Important Copyright Information

WARNING CONCERNING COPYRIGHT RESTRICTIONS

The copyright law of the United States (Title 17, United States Code) governs the making of photocopies or other reproductions of copyrighted materials.

Under certain conditions specified in the law, libraries and archives are authorized to furnish an article or other reproduction. One of these specified conditions is that the article or reproduction is not to be "used for any purpose other than private study, scholarship, or research". If a user makes a request for, or later uses, an article or reproduction for purposes in excess of "fair use", that user may be liable for copyright infringement.

This institution reserves the right to refuse to accept a copying order if, in its judgment, fulfillment of the order would involve violation of copyright law.



Resource Sharing Lending Slip Letter

11/14/2025

Journal Title: Annales geophysicae

Book/Article Title: Inversion of profiles of uplift rates for the geometry of dipslip faults at depth, with examples from the Alps and Himalaya

Author: MOLNAR P.

Chapter Title:

Volume: 5

Issue:

Pages:

Barcode: 50671007203539

Supplied To: Harvard University - Widener Library

Shipping Address:

Harvard University

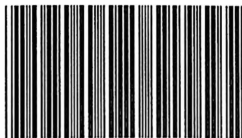
Widener Library

Library Email: delivery@library.gatech.edu

Borrower Reference:



My ID: 25643080



group_qualifier: 25643080

Format: DIGITAL

Date Needed By:

Inversion of profiles of uplift rates for the geometry of dip-slip faults at depth, with examples from the Alps and the Himalaya

P. MOLNAR (*)

*Laboratoire de Géophysique Interne et Tectonophysique, (U.A. 733), B.P. 68
Institut de Recherches Interdisciplinaires de Géologie et de Mécanique,
Université Scientifique Technologique et Médicale de Grenoble,
38402 Saint Martin d'Hères Cedex, France*

Received April 13, 1987 ; revised July 6, 1987 ; accepted July 10, 1987.

ABSTRACT. A simple method is presented for inverting a profile of uplift rates for the orientation and rate of slip on an underlying dip-slip fault, subject to the assumptions that deformation occurs only in the hanging wall and only by a simple shear on planes with a common orientation. An application to a profile of uplift rates across the Alps shows that uplift there could be due to underthrusting of the Eurasian plate beneath the Alps at a few (≈ 5) mm/y with a sense of shear within the Alps consistent with retrocharriage, or back-thrusting, of the Alps over the Po basin. An application to a profile of river terrace heights in the Himalaya, assuming them to be a measure of uplift rates, is consistent with a steepening of the main underthrust zone, the Main Boundary Fault, beneath the Greater Himalaya and therefore with that fault being a crustal scale ramp-overthrust.

Key words : uplift rates, thrust faulting, Alps, Himalaya.

Annales Geophysicae, 1987, **5B**, (6), 663-670.

INTRODUCTION

One of the difficulties faced by students of active tectonics is the determination of the styles and geometry of deformation occurring well below the surface of the earth. Seismicity, including the determination of fault plane solutions, provides invaluable information for seismically active areas, but not all faults and not all parts of some faults have been seismically active, at least during the intervals of time when local seismographs have been operated. As an example, one of the earth sciences' major controversies concerns the geometry of normal faulting ; a wave of enthusiasm for normal faulting on planes with very gentle dips ($\delta \leq 15^\circ$) has followed from the recognition of such features exhumed and exposed by several km of erosion (e.g., Davis *et al.*, 1980 ; Wernicke, 1981), but no well determined fault plane solution of a major earthquake ($M \geq 5.5$) has shown normal faulting on planes with dips significantly less than 30° (Jackson, 1987). It follows either that faults now dipping at low angles formed as steeper faults or that slip on active low-angle faults occurs aseismically.

Although less celebrated, the same controversy exists for thrust systems. Seeber *et al.* (1981) suggested that the great earthquakes in the Himalaya occur on very gently dipping thrust faults ($\delta \leq 5^\circ$) ; but fault plane solutions of many moderate earthquakes ($M = 5.5-6.5$) indicate dips greater than 15° , and those of some are $25^\circ-30^\circ$ (Baranowski *et al.*, 1984 ; Ni and Barazangi, 1984). Microearthquake activity does not define a gentle plane at all (Armbruster *et al.*, 1978 ; Gaur *et al.*, 1985). Similarly, if convergence continues in the Alps by underthrusting of the European platform beneath the range, then the geometry of faulting remains poorly constrained, for the seismicity of the Alps does not define a major underthrust zone. Thus, although the study of seismicity has been a vital tool for studying active tectonics of seismic regions, its users have had to be aware of the difficulties of interpreting the lack of earthquakes in a given area.

The geometry of faulting is reflected by surface deformation even when faults do not break the surface (Fitch and Scholz, 1971 ; Savage and Hastie, 1966), and therefore it should be possible to use profiles of geodetically or geologically determined uplift rates to constrain the geometry of faulting beneath them. The purpose of this paper is to develop this idea. Beginning from a formulation presented by White *et al.* (1986) for inverting the cross sectional profile of a warped bed for the geometry of an

(*) Permanent address : Department of Earth, Atmospheric, and Planetary Sciences, Massachusetts Institute of Technology, Cambridge, Mass. 02139, USA.

underlying fault, I present a slight addition and modification that allows profiles of uplift rates to be inverted for the geometry of the underlying fault.

When faults are not planar and when the direction of slip on them varies from place to place, internal deformation within at least one of the fault blocks must occur. Geodetically measured deformation, following a large earthquake, is usually assumed to be elastic and can be determined from the boundary conditions on the fault and on the earth's surface. For longer time-averaged, permanent deformation, however, strain will surely not be elastic. White *et al.* (1986) showed that if the strain occurs solely in the hanging wall by simple shear on planes of constant orientation, then the shape of a deformed bed (or any other surface of identical age) uniquely constrains the shape of the underlying fault. For regions of large-scale «*décollement*» in which a strong plate is underthrust beneath a range of mountains, the assumption that deformation is confined to the hanging wall probably is reasonable. Even if that deformation occurred largely by simple shear, however, the orientation of the shear would not necessarily be constant throughout the hanging wall. Nevertheless, one advantage of this formulation is that the effect of shear in the upper plate on planes of different orientations can be studied easily, and even when the style of deformation cannot be determined by other means, constraints on the geometry of faulting can be used to restrict the possible styles of deformation occurring in the hanging wall.

CALCULATION OF THE SHAPE OF A FAULT FROM A PROFILE OF MEASURED UPLIFT RATES

Consider a profile of a fault surface defined by $y(x)$, where x is the horizontal coordinate along the profile in the plane of the cross section (fig. 1a). Let us assume that the foot wall moves with a velocity $\mathbf{v} = (v_x, v_y)$ with respect to the undeforming part of the hanging wall, where

$$v_x = v \cdot \cos \Theta, \quad v_y = -v \cdot \sin \Theta$$

and Θ is the plunge of the slip vector (fig. 1a). For a curved fault, slip cannot occur without deformation of the «*upper plate*» or foot wall; let us assume that the foot wall does not deform. To start, let us also assume that slip occurs by simple shear solely on vertical planes.

Slip on the fault must occur parallel to the fault, but the vector describing the velocity of the foot wall need not be parallel to the fault. Consequently, two terms contribute to uplift at a point on the hanging wall. The first is simply the vertical component of velocity of the foot wall, v_y . The second is given by the product of the horizontal component of velocity and the slope of the fault, dy/dx , at the point in question. Thus the uplift rate, $u(x)$, can be written as

$$u(x) = v_y - v_x \cdot dy/dx = -v \cdot \sin \Theta - v \cdot \cos \Theta \cdot dy/dx.$$

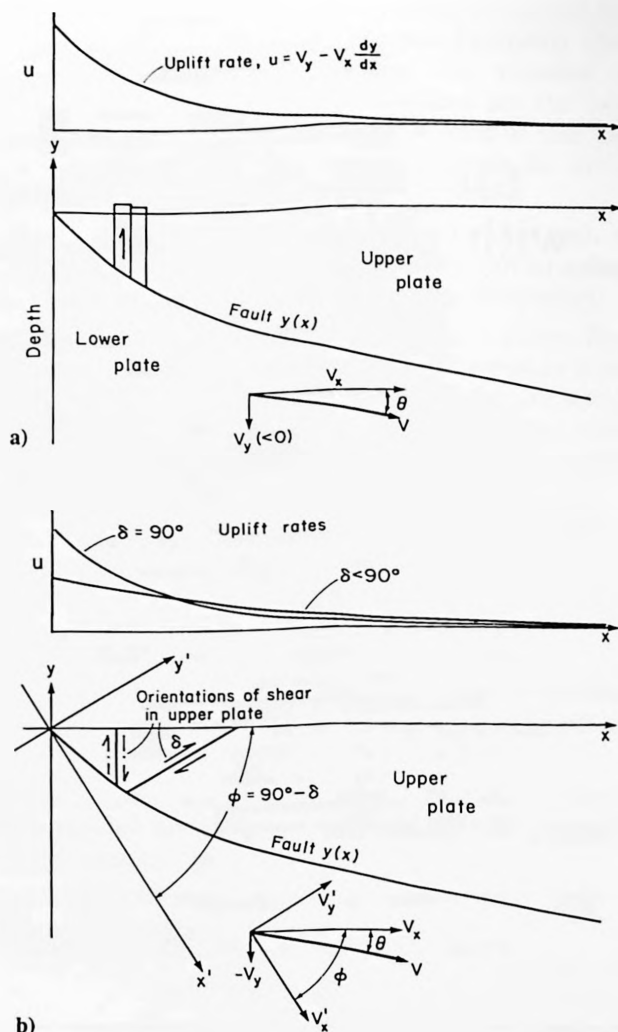


Figure 1

Geometry, coordinate system, and uplift rate, u , caused by slip on a curved fault, described by its depth ($y \leq 0$) as a function of horizontal position (x). Slip of the foot wall occurs at a velocity $\mathbf{v}$, plunging at an angle Θ beneath the hanging wall.

(a) Deformation in the hanging wall by simple shear on vertical planes yields the curve of uplift rate $u(x)$ shown above the coordinate system.

(b) If deformation occurs by simple shear on planes dipping at an angle δ , then to solve for $y(x)$, the coordinate system is rotated so that shear is parallel to the y' -axis. Note the effect of different assumed dips of shear zones in the hanging wall on the profiles of uplift rates.

Rearranging terms and integrating over x allows us to write $y(x)$ in terms of $u(x)$, v , and Θ :

$$y(x) - y_0 = - \int_{x_0}^x \left(\tan \Theta + \frac{u(x)}{v \cdot \cos \Theta} \right) dx$$

or

$$y(x) = y_0 - (x - x_0) \cdot \tan \Theta - \frac{1}{v \cdot \cos \Theta} \int_{x_0}^x u(x) dx \quad (1)$$

for which the fault is assumed to pass through $y(x_0) = y_0$. Thus given the depth of the fault ($-y_0$) at a particular locality (x_0), the speed and plunge of the foot wall, and a profile of uplift $u(x)$, one can solve for the profile of the fault, $y(x)$.

As an example consider an uplift profile given by

$$u(x) = u_0 \cdot x \cdot e^{1-x/d}/d \quad (2)$$

which reaches a maximum value of u_0 at $x = d$. Suppose that this uplift is due to convergence at a rate v_0 , for which the plunge, Θ , is zero. Let the fault crop out at the origin (0, 0). Thus, putting (2) into (1) yields

$$y(x) = \frac{u_0}{v_0} \cdot [(x + d) \cdot e^{1-x/d} - d \cdot e] \quad (3)$$

Note that if the maximum rate of uplift were 25 % of the convergence rate ($u_0 = 0.25 \cdot v_0$), and if the rate of uplift reached its maximum 40 km from the toe of the fault (at $d = 40$), then the maximum depth of the fault would be $(0.25 \cdot d \cdot e) = 27$ km. The dip would be zero both at the surface and where it approached its maximum depth, and it would be a maximum of 14° at $x = d = 40$ km where $dy/dx = u_0/v_0 = 0.25$. Clearly this fault is a ramp overthrust.

The same basic procedure can be followed if the hanging wall deforms by simple shear on planes dipping at an angle δ , where δ is positive for a sense of shear antithetic to that on the fault (fig. 1b). Following the approach of White *et al.* (1986), let us transform to a coordinate system x', y' in which simple shear occurs on plane parallel to the y' -axis (fig. 1b). Letting $\phi = 90^\circ - \delta$, we have

$$\begin{aligned} x' &= x \cdot \cos \phi - y \cdot \sin \phi \\ y' &= x \cdot \sin \phi + y \cdot \cos \phi \end{aligned}$$

We can relate v'_x and v'_y similarly to v_x and v_y , or by $v'_x = v \cdot \cos (\Theta - \phi)$ and $v'_y = -v \cdot \sin (\Theta - \phi)$.

The « uplift » in the y' direction

$$u' = u / \sin \delta = u / \cos \phi \quad (4)$$

again is due both to the y' -component of slip on the fault and to deformation of the hanging wall. Thus, in

the primed coordinate system we again have

$$u' = v'_y - v'_x \frac{dy'}{dx'}$$

and that

$$y'(x) = y'_0 - (x' - x'_0) \cdot \tan (\Theta - \phi) - \frac{1}{v'_x} \int_{x'_0}^{x'} u'(x') dx' \quad (5)$$

This expression yields a sequence of pairs of x' and y' that can be transformed into a series of pairs of x and y by

$$\begin{aligned} x &= x' \cdot \cos \phi + y' \cdot \sin \phi \\ y &= -x' \cdot \sin \phi + y' \cdot \cos \phi \end{aligned} \quad (6)$$

and therefore that define $y(x)$, the profile of the fault. Clearly for $\phi = 0$, (5) reduces to (3).

To illustrate the roles played by the velocity v and the plunge Θ of the foot wall, and of the dip δ of the shear zones, let us consider again the profile of uplift used in (2). When $\delta = 90^\circ$ and $\Theta = 0^\circ$, the inferred depth of the fault is inversely proportional to v , but if δ does not equal 90° , this proportionality is only approximate.

For instance, for $u_0 = 1$ mm/y and $v = 8, 4$, and 2 mm/y, the depths of the fault at $x = 48$ km are 10, 23, and 63 km (fig. 2a), clearly not in the ratios 1:2:4. Shear on increasingly flatter zones in the hanging wall is associated with steeper faults and with an increasing depth of the fault beneath any locality (fig. 2). If the foot wall plunges beneath the hanging wall at an angle Θ , then to cause uplift of the hanging wall, the fault must dip more steeply than Θ . Accordingly the greater is the assumed value of Θ , the greater will the inferred depth of the fault be (fig. 2b).

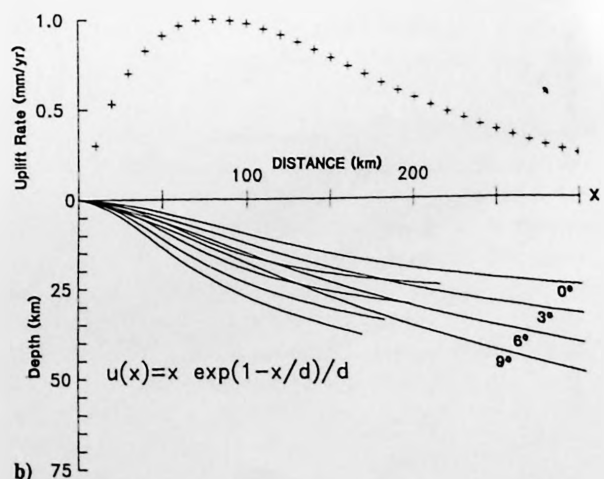
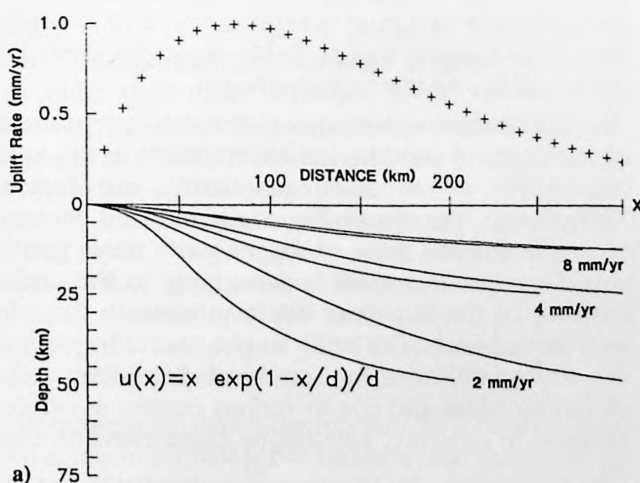


Figure 2

Plots of a hypothetical profile of uplift rate, $u(x)$, shown by plus signs, and several plausible inferred cross sectional profiles of faults for different assumed rates and plunges of underthrusting and for different assumed dips (δ) of shear zones in the hanging wall. The uplift rate is given by: $u(x) = x \cdot \exp(1 - x/d)/d$, where the maximum rate is 1 mm/y and $d = 40$ km.

(a) Inferred cross sectional shapes of faults assuming rates of underthrusting of 2, 4, and 8 mm/y and for shear on planes dipping vertically and at 45° . The lower of each pair of curves was calculated assuming shear on planes dipping at 45° , and the upper ones are for shear on vertical planes. Thus, for the slower rates of convergence, faults would be inferred to be deeper, and for the more gently dipping shear zones in the hanging wall, faults would be inferred to be steeper.

(b) Inferred cross sectional shapes of faults for different plunges of underthrusting ($0^\circ, 3^\circ, 6^\circ$, and 9°), for shear on planes dipping vertically and at 30° . As in (a), the steeper and shorter lines show the inferred shapes of faults for shear on the more gently dipping planes, in this case at 30° . For the greater assumed plunges, calculated dips of faults are steeper.

Although an increase in Θ or a decrease in v or in δ manifest themselves by greater inferred depths of the fault at any given position, x , variations in the quantities affect the inferred dips of the fault differently (fig. 2). Thus with sufficiently accurate data, it should be possible to distinguish them separately. In practice, however, we might expect to know v from plate motions or from geodetic data, and we might be able to determine the dip and depth of the fault from studies of earthquakes. In such we can use the profiles of uplift to deduce the other parameters.

APPLICATION TO THE ALPS

An ideal region to apply the formalism described above would be one for which a profile of uplift was well defined and beneath which there existed a single fault. To the best of my knowledge, direct measurements of average uplift rates, measured over any meaningful length of time, are rare, and profiles of uplift rates do not exist for any active chain except the Alps. Repeated leveling across the Alps in Switzerland shows the most rapid uplift to be south of the west of the range (Gubler *et al.*, 1981), and a north-northwest-south-southeast profile from Basel to Italy (fig. 3) shows a smooth increase across the Alps and a smooth decrease southward (Rybach *et al.*, 1980). The European platform north of the Alps has been flexed down beneath the Molasse basin and underthrust southward beneath the Alps (Karner and Watts, 1983). Whether this process continues, however, and whether or not a major active thrust fault dips south beneath the Alps do not seem to be known at present. Thus, the application of the formalism described above to the profile of uplift rates is predicated on an as yet untested assumption.

Let us choose an origin of the coordinate system at the

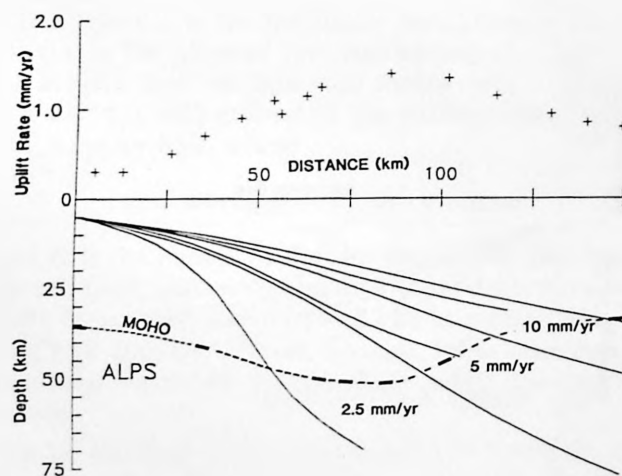


Figure 3

Profile of measured uplift rates, shown by plus signs, across the western Alps in Switzerland, from northwest (left) to southeast (right) (from Gubler *et al.*, 1981 and Rybach *et al.*, 1980), and inferred cross sectional profiles of faults for rates of underthrusting of 2.5, 5, and 10 mm/yr and for shear on planes dipping vertically and at 45°. For each pair of possible cross sectional shapes for the same rate of underthrusting, the steeper one is for shear on planes dipping 45°. The position of the Moho, inferred from seismic refraction profiles (Rybach *et al.*, 1980), is also shown. The dip and the depth of the fault at the edge of the range ($x = 0$) were taken to be 6° and 5 km, respectively.

edge of the Molasse basin, where the basement lies at a depth of ($-y_0 =$) 5 km and where the basement dips beneath the range at an angle of ($\Theta =$) 6° (Büchi and Trümpy, 1976; Rybach *et al.*, 1980). Since we do not know what the convergence rate is across the Alps or even if an active thrust fault dips southward beneath the Alps, we can use the measured uplift rates and the formation outlined above to ask the question: What could that convergence rate be if such a fault did exist? The extensive southeastward back-thrusting, or « retrocharriage », in the southern part of the Alps, which reflects shear of a thrust sense on planes dipping north or northwest, makes it reasonable to allow simple shear on planes with a range of northwards dips. To examine plausible ranges of parameters, we inverted the uplift rate profile using convergence rates of 10, 5, and 2.5 mm/yr and assuming dips of simple shear zones of 90° and 45° (fig. 3).

For all the convergence rates considered, if uplift in the southern part of the Alps occurs by overthrusting on a single plane with shear in the hanging wall on vertical planes, then the fault must penetrate into the mantle beneath the southern part of the Alps (fig. 3). The slightly upward curvature of a fault penetrating into the upper mantle seems unlikely, to me at least. Thus, insofar as the assumption of little deformation in the footwall is reasonable, either shear in the southern Alps occurs on planes dipping at a gentle angle or the uplift of the southern alps is not a consequence of slip on a single, southward dipping thrust fault. For shear on planes dipping 45° and for a convergence rate of only 2.5 mm/yr, the fault would reach the Moho north of the root of the Alps. If the processes that built the Alps are continuing, then this is improbable because the root surely formed by underthrusting of the European platform beneath the range. Thus insofar as it is reasonable to expect the present uplift of the Alps to be due to continued underthrusting of the European platform, convergence probably occurs at several mm/yr, and deformation of the hanging wall probably occurs by shear in a sense parallel to the retrocharriage.

Given the numerous untested assumptions, it probably is premature to consider the uncertainties in the uplift rate profile or to draw quantitative conclusions. Nevertheless, the plausibility of the inferred convergence rate and the sense of shear clearly allow for the possibility that this uplift is due simply to the underthrusting of the European Platform beneath the Alps on a curved fault. The uplift might, therefore, not be due to a complicated interaction of distributed strain of varying styles and due to various causes, as often is assumed in complex, non-unique finite-element models.

APPLICATION TO THE HIMALAYA

Unlike the Alps the evidence for a single major active thrust fault dipping beneath the range is quite convincing. The basement lies beneath about 4-5 km of Cenozoic sediment at the foot of the range and dips gently at 2°-3° northward there (Karunakaran and Ranga Rao, 1979; Sastri *et al.*, 1971). Fault plane

solutions and focal depths of earthquakes suggest that the top surface of the Indian shield lies at a depth of about 15 km some 80 to 100 km north of the foot of the range (Baranowski *et al.*, 1984; Ni and Barazangi, 1984). Finally gravity anomalies are consistent with a very gentle average dip ($\approx 3^\circ$) of the Moho beneath the Lesser Himalaya, but with a steeper dip beneath the Greater Himalaya (Lyon-Caen and Molnar, 1983, 1985). Moreover, some fault plane solutions imply more steeply dipping planes ($25\text{--}35^\circ$) beneath the Greater than Lesser Himalaya ($< 15^\circ$) (Baranowski *et al.*, 1984; Ni and Barazangi, 1984). Finally, the rate of convergence between the Indian Shield and the foot on the Himalaya is constrained to be 15 ± 5 mm/y (Lyon-Caen and Molnar, 1985). What we lack from the Himalaya for this study is a profile of uplift rates across the range.

In order to illustrate the use of the formalism described above, let us treat the profile of heights of river terraces along the Kali Gandaki in Nepal, given by Iwata *et al.* (1984), as a measure of the uplift rate. The Kali Gandaki, one of the major rivers crossing the Himalaya (fig. 4a), is bordered by a network of terraces along most of its length from the Tibetan Himalaya, north of the high peaks, to the plains of northern India (Fort *et al.*, 1981, 1982; Iwata, 1984; Iwata and Nakata, 1987; Iwata *et al.*, 1982, 1984). The highest terraces are as much as 300 to 500 m above the present river valley, and heights of lower terraces range from tens of meters to more than 100 m (fig. 4b). I have used the highest terraces as a measure of uplift, and for the purpose of the analysis described below, I assumed that the profile of the height of that terrace above the present river valley, given by Iwata *et al.* (1984), yields a profile of uplift since some, poorly known, time. This assumption, in fact, consists of several separate assumptions, each of which is difficult to justify to any quantitative degree.

First, there is a risk that the highest terrace did not form at the same time everywhere along the river valley. Since this terrace in the upper reaches of the Kali Gandaki contains morainal deposits that clearly are older than those deposited by the last glaciation, the terrace there is surely older than 50000 years. Iwata *et al.* (1982, 1984) inferred that it dates from the penultimate glacial maximum, some 150000 years ago. In any case, the age is too great for radiocarbon dating and too young for most other methods of dating. The main reason for my using the highest terrace is that it is the most continuous of those mapped by Iwata *et al.* (1984) and hence the least likely to be miscorrelated.

The process by which the terraces formed need not be the same in all cases. For instance, the highest terrace in the Thakhol segment (fig. 4), where the river reaches the Greater Himalaya, has been interpreted as a lake bed instead of an alluvial terrace (Fort *et al.*, 1981, 1982). The elevation of that terrace is nearly flat (Iwata, 1984; Iwata *et al.*, 1982), as it would be if it marked an undeformed old lake bed, and much of the material underlying the terraces is inferred to be lacustrine (Fort *et al.*, 1981, 1982; Iwata, 1984). Iwata *et al.* (1982), however, argued that many of the deposits in the terrace are stream cobbles. Moreover,

they pointed out that not only do the heights of all the younger terraces above the present valley increase downstream, but also the terrace profiles increase in slope with decreasing age. Thus, they inferred that the terrace profiles in the Greater Himalaya reflect a tilting down to the north (Iwata, 1984, Iwata *et al.*, 1984), a phenomenon inferred also by Fort (1980, Fort *et al.*, 1982) from the northward tilted beds of older material in the Thakola graben.

This increase in heights of terraces above the present valley is mirrored south of the Greater Himalaya, from Dana to south of where the Modi Khola and Kali Gandaki join (fig. 4); Yamanaka and Iwata (1982) found that the heights of low (30-70 m), intermediate (100-200 m), and high (250-400 m) terraces above the present river all decrease downstream in that segment. They interpreted this as an upward tilt to the north. Clearly these heights require greater incision upstream than downstream, but whether they also require uplift and tilting cannot be proven. There is one segment of the Kali Gandaki, however, where tilting is required; near Narayanghat, where this river crosses an anticlinal ridge of Tertiary (Siwalik) molasse, recent terraces are tilted northwards so much that elevations above sea level decrease upstream (northwards) (Iwata and Nakata, 1987). The same seems to exist north of Narayanghat across the Mahabharat Lekh (fig. 4) (Iwata *et al.*, 1984).

By treating the heights of the high terrace along the Kali Gandaki as a proportional measure of uplift, I implicitly assume that in some sense the river profile is in steady state and therefore that the Himalaya, carrying terraces that are assumed to be of the same age, has moved upward with respect to that profile. I do not like making these assumptions, and the reader probably will, and certainly should, treat the following analysis more as an example than as a proof of how tectonics occurs in the Himalaya. Nevertheless, I go forward, in part because the resulting inferred cross sectional shape of the fault is qualitatively consistent with my prejudice that the main Boundary Fault is a crustal scale ramp-overthrust (Lyon-Caen and Molnar, 1983).

For an origin, I assumed a depth of 5 km at the foot of the Mahabharat near Narayanghat, which is some 20 km north of the edge of the plains of India. I also took the dip θ of the fault there to be 2° , consistent with the dip of the basement beneath the plains (Karunakaran and Ranga Rao, 1979) and with gravity anomalies (Lyon-Caen and Molnar, 1983, 1985). I projected the profile given by Iwata *et al.* (1984), which was measured along the stream, onto a line with an azimuth of $N10^\circ E$, perpendicular to the local strike of the range. To convert this profile of elevation differences $h(x)$ or $h'(x')$ to a profile of uplift rates $u(x)$ or $u'(x')$ in (5), it is necessary to divide them by an assumed age for the terrace [$u'(x') = h'(x')/T$]. Calculations were made with that age chosen to be 150000 years, but notice from (5) that calculations of the shape of the fault depend on the product of that age and the convergence rate ($T \cdot v_0$). Thus, other ages can be treated simply by varying the convergence rate. Calculations are shown in figure 5 for rates of 10, 15, and 25 mm/y, but those for 10 mm/y correspond

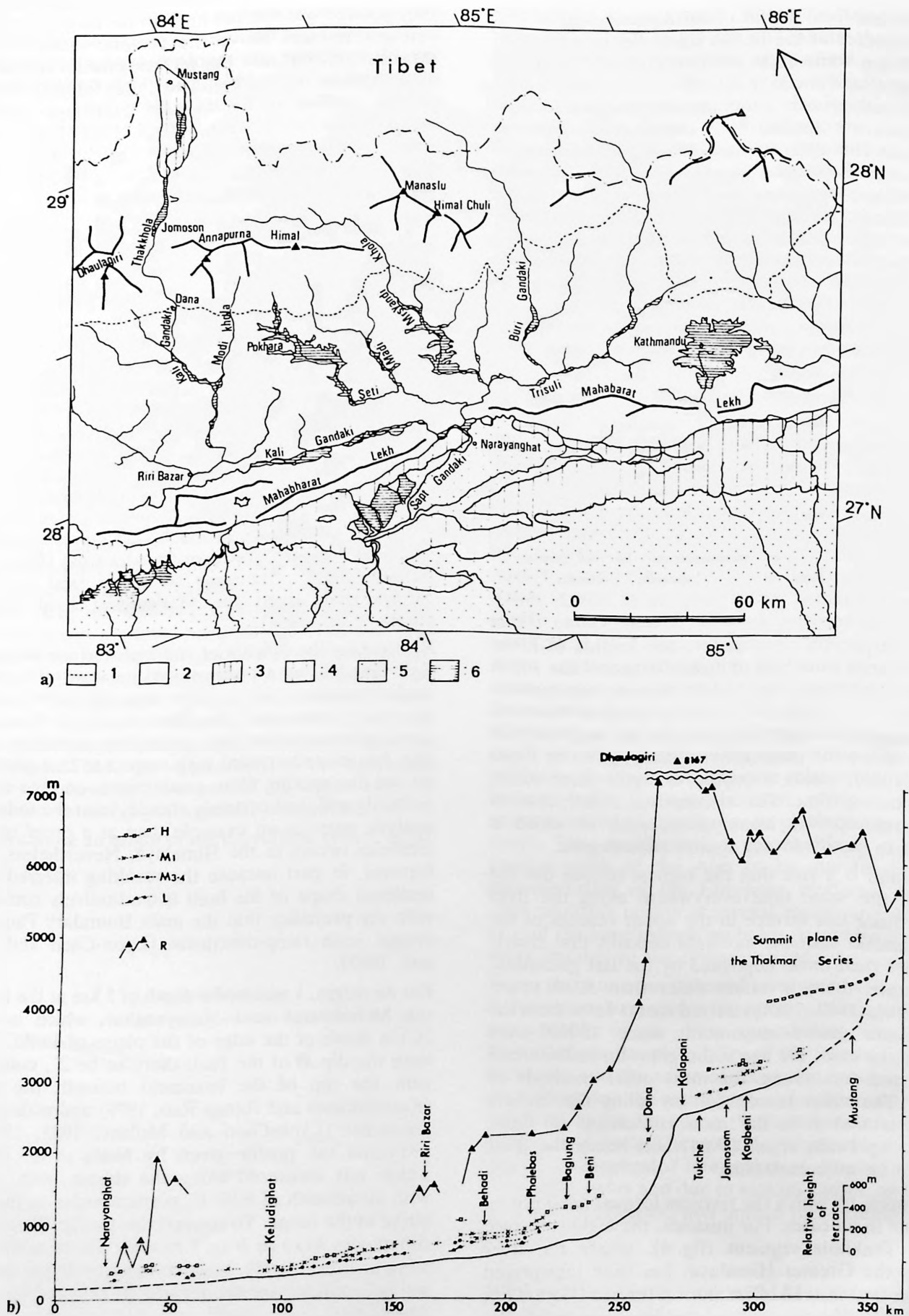


Figure 4

Map (a) of part of Nepal and longitudinal profile (b) of the Kali Gandaki and its terraces (from Iwata et al., 1984). The single dashed line across the center of the map marks the Main Central Thrust. Blank areas indicate pre-Tertiary bedrock. Closely spaced vertical hatching shows outcrops of the Thakmar series of late Cenozoic lacustrine and gravel deposits in the Thakholra graben. Widely spaced vertical hatching shows outcrops of the Siwalik sequence of late Cenozoic molasse. The dotted area is covered by young alluvium. Closely spaced horizontal hatching indicates outcrops of Pleistocene and Holocene terrace deposits. In profile (b), the different terraces correlated with one another are indicated by the same symbols; I have used the highest terraces, called H by Iwata et al. (1984). The neighboring topography is also shown, by dark triangles (R).

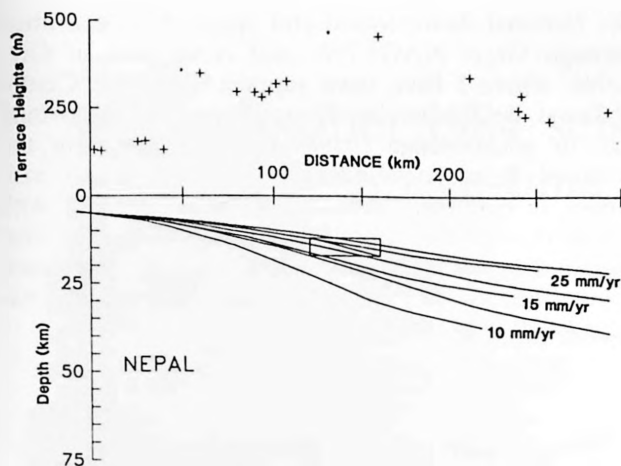


Figure 5

Inversion of terrace heights (H in fig. 4) projected onto a profile oriented $N10^\circ E$ and shown as plus signs, for possible cross sectional profiles of faults, shown below. The age of the terraces is assumed to be 150000 years. Inversions were made for rates of underthrusting of 10, 15, and 25 mm/y and for shear in the upper plate on planes dipping vertically and at 45° . For each pair of possible cross sectional shapes, the shorter and steeper profile is for shear on planes dipping at 45° , and the gentler one is for shear on vertical planes. The box in the middle shows where moderate earthquakes (with depths between 12 and 18 km) occur, and the diagonal of the box dips at 15° , the average dip inferred from fault plane solutions of such earthquakes (Baranowski et al., 1984; Ni and Barazangi, 1984). To be consistent with both the depths of these earthquakes and the dips of the fault planes, inferred faults should pass through this box with dips close to or steeper than 15° . The dip and the depth of the fault are assumed to be 2° and 5 km, respectively, at the foot of the Himalaya ($x = 0$).

also to ages of 100000 years and a rate of 15 mm/y or to 60000 years and a rate of 25 mm/y. Finally, also projected onto the inferred profiles of the fault is the region in which moderate earthquakes typically occur, the region for which we know focal depths and fault plane solutions (fig. 5). Insofar as the locations and fault plane solutions of earthquakes in the Himalaya can be projected onto the profile along the Kali Gandaki, any inferred fault should pass through this region and with a dip of about $15^\circ \pm 10^\circ$.

For a velocity of 10 mm/y, or for assumed ages of the terraces greater than 150000 years, the inferred profiles of the fault lie deeper than the depths of the earthquakes that are thought to occur on that fault (fig. 5). Thus insofar as the terraces can be used to measure the uplift rates, the rate of underthrusting is greater than 10 mm/y, or the terraces must be younger than 150000 years.

Note that all possible inferred fault profiles are steeper beneath the southern edge of the Greater Himalaya than beneath the Lesser Himalaya, an inference drawn also from gravity anomalies and from the warping of structures in the Himalaya (Lyon-Caen and Molnar, 1983, 1985). The calculated dips of the planes, however, are not larger than 15° . Thus if this profile of terrace heights could be used to infer representative uplift rates for the Himalaya, then those earthquakes for which fault plane solutions indicate planes dipping as steeply as 25° could not have occurred on the main underthrust zone. Accordingly, the dip of that fault shown in an idealized, but presumed to be representative, cross section through

the crust of the Himalaya by Molnar (1984) would be too steep. The uncertainty in the extent to which this profile of terrace heights is representative of uplift rates for the Himalaya, or even of this area, however, does not permit us to reject the possibility of the main overthrust dipping 25° to the north.

The maximum inferred dip of the fault is critically dependent upon the value of the maximum uplift rate, which probably is one of the least certain (see Iwata et al., 1982, 1984). For instance, if its value were twice as large as assumed in figure 5, the maximum dip of the fault could be 17° for shear on vertical planes and 23° for shear on planes dipping 45° , for a rate of underthrusting of 15 mm/y.

Given the doubts that must surround the quantitative application of the formalism outlined above to the measured terrace heights and their inferred ages, it probably is unwise to draw quantitative conclusions about the geometry of faulting. Nevertheless, the increase in such heights upstream near the foot of the Greater Himalaya (Iwata et al., 1982; Yamanaka and Iwata, 1982) does suggest that the fault beneath that area is steeper than it is farther south.

DISCUSSION AND CONCLUSION

Using the approach taken by White et al. (1986) to solve a related problem, a unique solution for the cross sectional shape of a dip-slip fault is given in terms of a profile of the uplift rates above the fault, the velocity of convergence, the plunge of that velocity, and the dip of zones of simple shear in the hanging wall, along which all of the deformation is assumed to occur. Clearly this method should not be applied to regions in which deformation of the foot wall is likely to be occurring, and the assumption that deformation in the hanging wall occurs by simple shear on planes of constant orientation will probably make inferences drawn from it only qualitative useful. Nevertheless, where a strong plate is underthrust beneath a thin sheet, or sheets, of sedimentary or crystalline rock, the uplift of the hanging wall is likely to depend strongly on the cross sectional shape of the fault, and this method should give a meaningful impression of the geometry of that fault.

An application to a profile of geodetically measured uplift rates across the Alps suggests that the uplift there could be the result of continued underthrusting of the European platform beneath the Alps at about 5 mm/y and with continuing retrocharriage in the southern Alps.

An application to heights of river terraces above the Kali Gandaki implies that the main underthrust zone steepens beneath the Greater Himalaya.

If the formalism outlined here is indeed applicable to convergent zones at the scale of 100's of km, then it may allow some quantification of the strains occurring within the hanging wall. For instance, although Lyon-Caen and Molnar (1985) argued that the convergence rate of the Indian plate and the Himalaya does not exceed 20 mm/y, they ignored possible strain within the Himalaya. Rapid uplift rates of a few mm/y could

be associated with convergent strain of comparable amounts (up to 5 mm/y of crustal shortening). Therefore the upper bound for the rate at which India moves towards southern Tibet probably should be increased to 25 mm/y. I suggest using 18 ± 7 mm/y.

Acknowledgements

I thank J. A. Jackson for a critical review of the manuscript. This research was supported in part by

the National Aeronautical and Space Administration through Grant NAG5-795, and completed in Grenoble, where I have been supported by the Centre National de Recherches Scientifiques. I also would like to acknowledge that a travel grant from the National Science Foundation allowed me to visit Nepal, in February, 1986, and become familiar with the work of S. Iwata and his colleagues and to visit Britain where I talked with Jackson and D. McKenzie about their work with N. White. My study was stimulated by those two visits.

REFERENCES

- Armbruster J., Seeber L., Jacob K. H., 1978. The northwestern termination of the Himalayan mountain front: Active tectonics from microearthquakes. *J. Geophys. Res.*, **83**, 269-281.
- Baranowski J., Armbruster J., Seeber L., Molnar P., 1984. Focal depths and fault plane solutions of earthquakes and active tectonics of the Himalaya. *J. Geophys. Res.*, **89**, 6918-6928.
- Büchi U. P., Trümpy R., 1976. Bemerkungen zum geologischen profil längs der geotransverse Basel-Chiasso. *Schweiz. Mineral. Petrogr. Mitt.*, **56**, 589-603.
- Davis G. A., Anderson J. L., Frost E. G., Shackelford T. J., 1980. Mylonitization and detachment faulting in the Whipple-Buckskin-Rawhide Mountains terrain, southeastern California and western Arizona. In: Cordilleran Metamorphic Complexes, *Geol. Soc. Am. Mem.*, **153**, 79-130.
- Fitch T. J., Scholz C. H., 1971. Mechanism of underthrusting in southwest Japan: A model of convergent plate interaction. *J. Geophys. Res.*, **76**, 7260-7292.
- Fort M., 1980. La formation de la Kali Gandaki et l'histoire récente de la haute chaîne himalayenne. *Bull. Assoc. Géogr. France*, **471**, 237-247.
- Fort M., Freytet P., Colchen M., 1982. Structural and sedimentological evolution of the Thakkhola Mustang graben (Nepal Himalayas). *Z. Geomorph. N. F.*, **42**, 75-98.
- Fort M., Bassoullet J. P., Colchen M., Freytet P., 1981. Sedimentological and structural evolution of the Thakkhola-Mustang graben (Nepal Himalaya) during late Neogene and Pleistocene. In: Proc. of Neogene/Quaternary Boundary Field Conf., India, 25-35.
- Gaur V. K., Chander R., Sarkar I., Khattri K. N., Sinval H., 1985. Seismicity and the state of stress from investigations of local earthquakes in the Kumaon Himalaya. *Tectonophysics*, **118**, 243-251.
- Gubler E., Kahle H.-G., Klingelé E., Mueller St., Olivier R., 1981. Recent crustal movements in Switzerland and their geophysical interpretation. *Tectonophysics*, **71**, 125-152.
- Iwata S., 1984. Geomorphology of the Thakkhola-Mukthinath region, central Nepal, and its late Quaternary history. Geograph. rep., Tokyo Metropol. Univ., **19**, 25-42.
- Iwata S., Nakata T., 1987. River terraces and crustal movement around Narayanghat, central Nepal. *J. Nepal Geol. Soc.* (in press).
- Iwata S., Yamanaka H., Yoshida M., 1982. Glacial landforms and river terraces in the Thakkhola region, central Nepal. *J. Nepal Geol. Soc.*, **2**, 81-94.
- Iwata S., Sharma T., Yamanaka H., 1984. A preliminary report on geomorphology of central Nepal and Himalayan uplift. *J. Nepal Geol. Soc.*, **4**, 141-149.
- Jackson J. A., 1987. Active normal faulting and continental extension. In: Continental Extensional Tectonics, Spec. Publ. *Geol. Soc. Lond.* (in press).
- Jackson J. A., Fitch T. J., 1981. Basement faulting and the focal depths of the larger earthquakes in the Zagros Mountains (Iran). *Geophys. J. Roy. Astron. Soc.*, **64**, 561-586.
- Karukaranan C., Ranga Rao A., 1979. Status of exploration for hydrocarbons in the Himalayan region. Contribution to Stratigraphy and Structure, Himalayan Geology Seminar, Section III, Oil and Natural Gas Resources, *Geol. Surv. Ind. Misc. Publ.*, **41**, 1-66.
- Karner G. D., Watts A. B., 1983. Gravity anomalies and flexure of the lithosphere at mountain ranges. *J. Geophys. Res.*, **88**, 10449-10477.
- Lyon-Caen H., Molnar P., 1983. Constraints on the structure of the Himalaya from an analysis of gravity anomalies and a flexural model of the lithosphere. *J. Geophys. Res.*, **88**, 8171-8191.
- Lyon-Caen H., Molnar P., 1985. Gravity anomalies, flexure of the Indian plate, and the structure, support and evolution of the Himalaya and Ganga Basin. *Tectonics*, **4**, 513-538.
- Molnar P., 1984. Structure and tectonics of the Himalaya: Constraints and implications of geophysical data. *Ann. Rev. Earth Planet. Sci.*, **12**, 489-518.
- Ni J., Barazangi M., 1984. Seismotectonics of the Himalayan collision zone: Geometry of the underthrusting Indian plate beneath the Himalaya. *J. Geophys. Res.*, **89**, 1147-1163.
- Rybach L., Mueller St., Milnes A. G., Ansorge J., Bernoulli D., Frey M., 1980. The Swiss geotransverse Basel-Chiasso — a review, *Eclogae Geol. Helv.*, **73/2**, 437-462.
- Sastri V. V., Bhandari L. L., Raju A. T. R., Datta A. K., 1971. Tectonic framework and subsurface stratigraphy of the Ganga basin. *J. Geol. Soc. India*, **12**, 223-233.
- Savage J. C., Hastie L. M., 1966. Surface deformation associated with dip-slip faultings. *J. Geophys. Res.*, **71**, 4897-4904.
- Seeber L., Armbruster J., Quittmeyer R., 1981. Seismicity and continental subduction in the Himalayan arc. In: Zagros, Hindukush, Himalaya, eds. H. K. Gupta and F. N. Delaney, *Geodyn. Ser.* **3**, A.G.U., Washington D.C., 215-242.
- Wernicke B., 1981. Low-angle normal faults in the Basin and Range Province-nappe tectonics in an extending orogen. *Nature*, **291**, 645-648.
- White N. J., Jackson J. A., McKenzie D. P., 1986. The relationship between the geometry of normal faults and that of sedimentary layers in their hanging walls. *J. Struct. Geol.*, **8**, 897-909.
- Yamanaka H., Iwata S., 1982. River terraces along the middle Kali Gandaki and Marsyandi Khola, central Nepal. *J. Nepal Geol. Soc.*, **2**, 95-112.